

Documentation for the symplectic methods

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This document expands on the comments at the beginning of
`integrators/symplectic_runge_kutta_nyström_integrator.hpp`.

1 Differential equations.

Recall that the equations solved by this class are

$$(q, p)' = X(q, p, t) = A(q, p) + B(q, p, t) \quad \begin{array}{l} \text{with } \exp hA \text{ and } \exp hB \text{ known} \\ \text{and } [B, [B, [B, A]]] = 0; \end{array} \quad (1.1)$$

$$\begin{array}{l} \text{the above equation, with } \exp hA = \mathbb{1} + hA, \exp hB = \mathbb{1} + hB, \\ \text{and } A \text{ and } B \text{ known;} \end{array} \quad (1.2)$$

$$q'' = -M^{-1} \nabla_q V(q, t). \quad (1.3)$$

2 Relation to Hamiltonian mechanics.

The third equation above is a reformulation of Hamilton's equations with a Hamiltonian of the form

$$H(q, p, t) = \frac{1}{2} p^\top M^{-1} p + V(q, t), \quad (2.1)$$

where $p = Mq'$.

3 A remark on non-autonomy.

Most treatments of these integrators write these differential equations as well as the corresponding Hamiltonian in an autonomous version, thus $X = A(q, p) + B(q, p)$ and $H(q, p, t) = \frac{1}{2} p^\top M^{-1} p + V(q)$. It is however possible to incorporate time, by considering it as an additional variable:

$$(q, p, t)' = X(q, p, t) = (A(q, p), 1) + (B(q, p, t), 0).$$

For equations of the form (1.3) it remains to be shown that Hamilton's equations with quadratic kinetic energy and a time-dependent potential satisfy $[B, [B, [B, A]]] = 0$. We introduce t and its conjugate momentum ϖ to the phase space, and write

$$\tilde{q} = (q, t), \quad \tilde{p} = (p, \varpi), \quad L(\tilde{p}) = \frac{1}{2} p^\top M^{-1} p + \varpi.$$

(1.3) follows from Hamilton's equations with

$$H(\tilde{q}, \tilde{p}) = L(\tilde{p}) + V(\tilde{q}) = \frac{1}{2} p^\top M^{-1} p + \varpi + V(q, t)$$

since we then get $t' = 1$. The desired property follows from the following lemma:

Lemma. *Let $L(\tilde{q}, \tilde{p})$ be a quadratic polynomial in $\tilde{p}$, $V(\tilde{q})$ a smooth function, $A = \{\cdot, L\}$, and $B = \{\cdot, V\}$. Then*

$$[B, [B, [B, A]]] = 0. \quad \square$$

Proof. It suffices to show that $\{V, \{V, \{L, V\}\}\} = 0$. It is immediate that every term in that expression will contain a third order partial derivative in the $\tilde{p}_i$ of L , and since L is quadratic in $\tilde{p}$ all such derivatives vanish. $\square$

See [MQo6, p. 26] for a detailed treatment of non-autonomous Hamiltonians using an extended phase space. See [McL93, p. 8] for a proof that $\{V, \{V, \{L, V\}\}\} = 0$ for arbitrary Poisson tensors.

4 Composition and first-same-as-last property

Recall from the comments that each step is computed as

$$(\mathbf{q}_{n+1}, \mathbf{p}_{n+1}) = \exp a_{r-1} h \mathbf{A} \exp b_{r-1} h \mathbf{B} \cdots \exp a_0 h \mathbf{A} \exp b_0 h \mathbf{B}(\mathbf{q}_n, \mathbf{p}_n),$$

thus, when b_0 vanishes (type ABA) or when a_{r-1} does (type BAB),

$$(\mathbf{q}_{n+1}, \mathbf{p}_{n+1}) = \exp a_{r-1} h \mathbf{A} \exp b_{r-1} h \mathbf{B} \cdots \exp b_1 h \mathbf{B} \exp a_0 h \mathbf{A}(\mathbf{q}_n, \mathbf{p}_n), \text{ respectively}$$

$$(\mathbf{q}_{n+1}, \mathbf{p}_{n+1}) = \exp b_{r-1} h \mathbf{B} \exp a_{r-2} h \mathbf{A} \cdots \exp a_0 h \mathbf{A} \exp b_0 h \mathbf{B}(\mathbf{q}_n, \mathbf{p}_n).$$

This leads to performance savings.

Let us consider a method of type BAB . Evidently, the evaluation of $\exp a_0 h \mathbf{A}$ is not required, thus only $r - 1$ evaluations of $\exp \Delta t \mathbf{A}$ are required. Furthermore, if output is not needed at step n , the computation of the $(n - 1)$ th step requires only $r - 1$ evaluations of $\exp \Delta t \mathbf{B}$, since the consecutive evaluations of $\exp b_0 h \mathbf{B}$ and $\exp b_r h \mathbf{B}$ can be merged by the group property,

$$\exp b_0 h \mathbf{B} \exp b_r h \mathbf{B} = \exp(b_0 + b_r) h \mathbf{B}.$$

If the equation is of the form 1.2, the latter saving can be achieved even for dense output, since only one evaluation of $\mathbf{B}$ is needed to compute the increments $b_r h \mathbf{B}$ and $b_0 h \mathbf{B}$.

The same arguments apply to type ABA . This motivates the name of the template parameter evaluations, equal to $r - 1$ for methods of type ABA and BAB , and r otherwise.

References

- [McL93] R. I. McLachlan. “Symplectic integration of Hamiltonian wave equations”. In: *Numerische Mathematik* 66.1 (1993), pp. 465–492. DOI: 10.1007/BF01385708.
- [MQo6] R. I. McLachlan and G. R. W. Quispel. “Geometric Integrators for ODEs”. In: *Journal of Physics A* 39 (2006), pp. 5251–5285. DOI: 10.1088/0305-4470/39/19/S01.